

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA0301	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	Total Marks:	100 Marks
CO1 Statement:	Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	SIOT D	Date:	23-07-2026

Code of Conduct
I D. Pujitha (192373037) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: D. Pujitha.

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

95
100

ASSESSMENT TOOL-2
SHORT ANSWER

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COURSE NAME :- MOBILE COMPUTING

- 1) Total spectrum (KHz) = spectrum (MHz)
Full-duplex channel bandwidth = $2 \times$ simple channel bandwidth
Total spectrum (KHz) = $45 \text{ MHz} = 45,000 \text{ KHz}$
 $= 2 \times 25 = 50 \text{ KHz}$
No. of full duplex channels = Total spectrum / Full-duplex channel bandwidth
 $= 45,000 / 50 = 900$
- 2) channels per cell = Total voice channels / cluster size (N)
 $= 560 / 7 = 80$ per cell.
- 3) Total voice channels = No. of carriers $\times$ Time slots per carrier
 $= 200 \times 4 = 800$ channels
- 4) Power ratio (dB) = $10 \times \log_{10} (P_1/P_2)$
 P_1 = Base station, P_2 = Handset power
Power ratio = $80 \div 4 = 20$
 $= 10 \times \log_{10}(20) = 10 \times 1.3010 = 13.01 \text{ dB} \approx 13 \text{ dB}$
- 5) $CI(\text{dB}) = \text{carrier channel (dBm)} - \text{Interference signal (dBm)}$
 $= (-60) - (-92) = 32 \text{ dB}$
- 6) Total channels = Total spectrum (MHz) / channel bandwidth (KHz)
 $= 42 \text{ MHz} = 42,000 \text{ KHz}$
 $\Rightarrow 42,000 / 300 = 140$
- 7) Received power $\propto 1/d^n$, new received power = original power / 2^n
 n = path-loss exponent.

Original power = 64W

distance double $\rightarrow$ Path loss factor = $2^4 = 16$, new power =

$$64 \div 16 = 4W$$

8) Required power increase = (new distance / original distance)ⁿ

n = 2 (Inverse-square law)

distance doubles

power increases = $2^2 = 4$, 4x increase in transmit power.

9) Total channels in cluster = cluster size $\times$ channels per cell
 $= 4 \times 15 = 60$

10) Doppler shift (f_d) = $(v \times f) / c$

v = Speed (m/s), f = carrier frequency (Hz), c = Speed of light = 3×10^8 m/s

Speed = 100 km/h = 27.78 m/s

$$\text{frequency} = 900 \text{ MHz} = 900 \times 10^6 \text{ Hz} \Rightarrow f_d = \frac{27.78 \times 900 \times 10^6}{3 \times 10^8} = 83.34 \text{ Hz}$$

11) Slot duration = frame duration / No. of time slots
 $= 5 / 8 = 0.625 \text{ ms}$

12) No. of carriers = Total spectrum / Carrier spacing $\Rightarrow 24000 / 200 = 120$ carriers

13) Transmit power (dBm) = $10 \log_{10}(P(\text{mW}))$

Received power (dBm) = Transmit power - path loss (dB)

Transmit power = 8W = 8000 mW, $P_{Tx} = 10 \log_{10}(8000) = 39.03 \text{ dBm}$

$$P_{Rx} = 39.03 - 33 = 6.03 \text{ dBm} \approx 6 \text{ dBm}$$

14) FSPL (dB) = $32.44 + 20 \log_{10}(d \text{ km}) + 20 \log_{10}(f \text{ MHz})$

d = distance (km), f = frequency (MHz)

$$= 32.44 + 20 \log_{10}(2.5) + 20 \log_{10}(1900) = 105.91 \text{ dB} \approx 106 \text{ dB}$$

15) Total capacity = No. of cells $\times$ calls per cell

$$= 6 \times 250 = 1500 \text{ capacity}$$

Active calls = call arrival rate $\times$ Avg call duration

$$\text{call arrival} = 4000$$

$$\text{call duration} = 180$$

$$\text{Active calls} = 4000 \times 180 = 720000$$

17) Total Bandwidth = No. of channels $\times$ Bandwidth per channel

$$= 150 \times 25 = 3750 \text{ KHz}$$

$$= 3.75 \text{ MHz}$$

18) No. of guard bands = No. of Bands - 1

Wasted bandwidth = No. of guard bands $\times$ guard bandwidth

$$\text{No. of bands} = 30$$

$$\text{Guard bands} = 30 - 1 = 29$$

$$\Rightarrow 29 \times 40 = 1160 \text{ KHz}$$

$$\Rightarrow 1.16 \text{ MHz}$$

19) cell diameter = $2 \times$ cell radius

$$\text{Time per cell} = \text{cell diameter} / \text{speed}$$

$$\text{Speed} = 90 \text{ km/h} = 25 \text{ m/s}$$

$$\text{Radius} = 1.5 \text{ km} = 1500 \text{ m}$$

$$\text{Diameter} = 3000 \text{ m}$$

$$\text{Time per cell} = 3000 / 25 = 120 \text{ s}$$

$$\text{Handoffs per hour} = 3600 / \text{Time per cell}$$

$$= 3600 / 120$$

$$= 30$$

$$20) \text{ Total users} = \text{Regional centers} \times \text{Base stations} / \text{center} \times \text{users / Base stations}$$

$$= 8 \times 40 \times 2500$$

$$= 80000$$

$$21) \text{ Approximate cell area} = R^2$$

$$\text{No. of cells} = \text{Total area} / \text{cell area}$$

$$\text{Radius} = 0.4 \text{ km}$$

$$\text{cell area} = (0.4)^2 = 0.16 \text{ km}^2$$

$$\text{cells} = 400 / 0.16 = 2500 \text{ cells}$$

$$22) \text{ Channels} = \text{Total spectrum} / \text{channel width}$$

$$= 6000 / 30 = 200$$

$$23) \text{ Traffic (Erlangs)} = (\text{subscribers} \times \text{calls / hour}) \times \text{Avg cell duration}$$

$$= 12000 / 60$$

$$= 200$$

$$24) \text{ Max users} = \text{Total bandwidth} / \text{Bandwidth per user}$$

$$= 40000 / 400$$

$$= 100$$

$$25) \text{ Total capacity} = \text{Users per base station} \times \text{No. of Base Stations}$$

$$= 600 \times 1200$$

$$= 720000$$

lagging operations : total calls / calls per lagged per operation

$$= 800/4 = 200$$

2a) offered traffic : arrival rate $\times$ avg call duration

$$= 40000 \times 3 = 120000$$

Blocked fraction : $(\text{offered traffic} - \text{system capacity}) / \text{offered traffic}$

$$= (120000 - 100000) / 120000$$

$$= 20000 / 120000 = 0.1667 \approx 0.17$$

$\Rightarrow 17\%$

2b) Roaming users : $(\text{Roaming pr} / 100) \times \text{Total subscribers}$

$$= 0.15 \times 600000$$

$$= 90000$$

2c) Utilization (%) : $(\text{current load} / \text{max capacity}) \times 100$

$$= 9000 / 12000 \times 100$$

$$= 75\%$$

3a) Total channels : $\text{Total spectrum} / \text{channel width}$

$$= 48000 / 800 = 60$$

3b) channels : $\text{Total spectrum} / \text{channel width}$

$$= 30000 / 200 = 150$$

3c) Successful Handoffs : $\text{Success rate} \times \text{Total Handoff Attempts}$

$$= 0.49 \times 2500$$

$$= 1225$$

33) Transmit power = $10 \log_{10} (P(\text{mW}))$
 Received power (dBm) = Transmit power - path loss
 $= 40.79 - 45$
 $= -4.21 \text{ dBm}$

$= 12 \text{ W}$
 $= 12000 \text{ mW}$
 $= 10 \log_{10} (2000)$
 $= 40.79 \text{ dBm}$

34) Total capacity = users per cell $\times$ No. of cells
 $= 900 \times 7 = 6300$

35) Slot duration = frame duration / No. of slots
 $= 4.8 / 8 = 0.6 \text{ ms}$

36) Doppler Shift (f_d) = $(v \times f) / c$
 v = Speed (m/s)
 f = carrier frequency (Hz), $c = 3 \times 10^8 \text{ m/s}$

Speed = $150 \text{ km/h} = 41.67 \text{ m/s}$

Frequency = $1.8 \text{ GHz} = 1.8 \times 10^9 \text{ Hz}$

$f_d = \frac{41.67 \times 1.8 \times 10^9}{3 \times 10^8} = 250 \text{ Hz}$

37) Offered traffic = arrival rate $\times$ avg call duration

Blocked fraction = $(\text{offered traffic} - \text{capacity}) / \text{offered traffic}$

offered traffic = $70000 \times 4 = 280000$

Blocked traffic = $280000 - 250000 = 30000$

Blocked fraction = $30000 / 280000 = 0.107 \approx 0.11 (11\%)$

$$\begin{aligned}\text{Total users} &= \text{users per cell} \times \text{No. of cells} \\ &= 600 \times 9 = 5400 \text{ users}\end{aligned}$$

$$\begin{aligned}39) \text{ Dropped calls} &= (\text{Drop percentage} / 100) \times \text{Total calls} \\ &= 0.004 \times 60000 \\ &= 240 \text{ calls}\end{aligned}$$

$$\begin{aligned}40) \text{ Path loss (dB)} &= \text{Transmit power (dBm)} - \text{Received power (dBm)} \\ &= 43 - (-87) \\ &= 130 \text{ dB}\end{aligned}$$

$$\begin{aligned}41) \text{ Total coverage} &= \text{coverage per base station} \times \text{No. of base station} \\ &= 2.5 \times 150 = 375 \text{ km}^2\end{aligned}$$

$$\begin{aligned}42) \text{ Total channels} &= \text{Total spectrum} / \text{channel width} \\ \text{channels per cell} &= \text{Total channels} / \text{frequency reuse factor} \\ \text{Total channels} &= 48000 / 200 = 240 \\ \text{channel per cell} &= 240 / 6 = 40\end{aligned}$$

$$\begin{aligned}43) \text{ Paging cycles} &= \text{Total users} / \text{users per cycle} \\ &= 150000 / 6000 = 25\end{aligned}$$

$$\begin{aligned}44) \text{ Handoffs} &= \text{call duration} / \text{handoff interval} \\ &= 9 / 3 = 3\end{aligned}$$

$$\begin{aligned}45) \text{ SNR (dB)} &= \text{signal power (dBm)} - \text{noise power (dBm)} \\ &= -85 - (-115) \\ &= 30 \text{ dB}\end{aligned}$$

46) Min Bandwidth = no. of users $\times$ data rate per user

$$= 250 \times 150$$

$$= 37500 \text{ Kbps} = 37.5 \text{ Mbps}$$

47) Traffic = (subscribers $\times$ calls per hour $\times$ avg call duration in min) $\div 60$

$$= 15000 \times 4 \times 0.5 = 15000 \text{ call} \cdot \text{min/hour}$$

$$= 15000 / 60 = 250 \text{ Erlangs}$$

48) Total bandwidth = no. of channels $\times$ Bandwidth per channel

$$= 250 \times 12$$

$$= 3000 \text{ KHz} = 3 \text{ MHz}$$

49) Total throughput = no. of connections $\times$ data rate per connection

$$= 1500 \times 600 = 900000 \text{ Kbps}$$

$$= 900 \text{ Mbps}$$

50) Area per Tower = πR^2

no. of towers = Total coverage area $\div$ Area per tower

$$\text{Area per Tower} = \pi R^2 = 3.14 \times 16 \approx 50.27 \text{ km}^2$$

$$\text{towers} = 350 / 50.27 \approx 6.96 \approx 7 \text{ towers}$$

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